RF_* COMMUTES WITH DERIVED LIMITS

THE STACKS PROJECT AUTHORS

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Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Derived products. Injectives, Lemma injectives-lemma-derived-products. Products in the derived category and their computation.
- Right adjoints preserve products. Categories, Lemma categories-lemma-adjoint-exact. Rf_* commutes with products.

Source reference. Original reference 07D9: sources/stacks-chow/tags.txt, source lines 9404–9404. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 079W: sources/stacks-chow/tags.txt, source lines 9286–9286. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0038: sources/stacks-chow/tags.txt, source lines 125–125. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Then Rf_* commutes with $R\lim$, i.e., Rf_* commutes with derived limits.*

Proof. Let (K_n) be an inverse system in $D(\mathcal{O}_X)$. Consider the defining distinguished triangle

$$R\lim K_n \to \prod K_n \to \prod K_n$$

in $D(\mathcal{O}_X)$. Applying the exact functor Rf_* we obtain the distinguished triangle

$$Rf_*(R\lim K_n) \to Rf_*\left(\prod K_n\right) \to Rf_*\left(\prod K_n\right)$$

in $D(\mathcal{O}_Y)$. Thus we see that it suffices to prove that Rf_* commutes with products in the derived category (which are not just given by products of complexes, see Injectives, Lemma 07D9). However, since Rf_* is a right adjoint by Lemma 079W this follows formally (see Categories, Lemma 0038). Caution: Note that we cannot apply Categories, Lemma 0038 directly as $R\lim K_n$ is not a limit in $D(\mathcal{O}_X)$. □

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